

ADI KARTHIKEYA SUNKARA BALASUBRAMANYAM

Houston, TX, USA · +1-832-654-1709 · adikarthikeya1234@gmail.com

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EXPERIENCE

Jan 2025 - May 2026 **RESEARCH ASSISTANT, MACHINE LEARNING AND AI** **HOUSTON, TX**
University of Houston (Petroleum Industry Partnership)

- **Subsurface ML:** Delivered subsurface-analytics models on Databricks and SQL Server, turning large-scale well and survey data into resource-optimization inputs the research team relied on for site decisions.
- **Pipeline and RAG:** Automated data-validation and model-evaluation pipelines in Python, cutting evaluation turnaround 40%, and stood up a RAG service over technical documentation (ChromaDB, FastAPI, LLMs) so the team could query specifications in plain language.

May 2025 - Jun 2025 **DATA SCIENCE INTERN, GEOSPATIAL ML** **HOUSTON, TX**
NSF Data Science for Energy Transition

- **Facies Classification:** Trained and tuned a facies-classification pipeline (PCA, Random Forest, XGBoost) on 50,000+ survey records, reaching 80.24% accuracy, 7 points above the logistic-regression baseline.
- **Visualization:** Put model predictions on candidate energy-transition sites in front of NSF stakeholders through Tableau and QGIS dashboards, and rewrote slow joins in pandas/NumPy to cut pipeline runtime 35%.

Apr 2023 - May 2024 **DATA ANALYST, ETL AND BI** **BENGALURU, INDIA**
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- **ETL Pipelines:** Developed Python and SQL ETL pipelines moving 1M+ records from internal sources into Azure Data Lake and downstream warehouses, cutting runtime 60% and stabilizing weekly reporting.
- **Modeling and BI:** Trained XGBoost and Random Forest segmentation models (89% accuracy, 0.91 ROC-AUC, 12% above the prior baseline), and shipped Power BI dashboards tracking 8 KPIs across three business units, adopted weekly by leadership.

TECHNICAL SKILLS

Languages: Python, SQL (PostgreSQL, MySQL, SQL Server, BigQuery), R, C

ML and AI: PyTorch, TensorFlow, HuggingFace, scikit-learn, XGBoost, LightGBM, SHAP, MLflow, LangChain, LangGraph, Google ADK, RAG, ChromaDB, Neo4j, SBERT, Gemini, Claude, OpenAI APIs

Cloud and MLOps: Azure Data Lake, Databricks, GCP (Cloud Run, BigQuery, Vertex AI), AWS, Docker, CI/CD, Git, GitHub, Linux

Backend and Visualization: FastAPI, Flask, REST APIs, Streamlit, Power BI, Tableau, pandas, NumPy

PROJECTS

2026 **RAREDX, AGENTIC RARE-DISEASE DIAGNOSTIC ASSISTANT** **LangGraph, Neo4j**

- **Diagnostic Workflow:** Engineered an agentic diagnostic workflow in LangGraph that pairs Neo4j graph retrieval with ChromaDB semantic search over Orphanet disease profiles, lifting Top-1 accuracy from 10% to 92.5% and Recall@5 from 22.5% to 100% on a 40-case benchmark.
- **Evaluation:** Ran ablations across 7 retrieval strategies (280 evaluations), calibrated confidence (Brier 0.068, ECE 0.051), and enforced PubMed-grounded outputs for a 0% hallucination rate.

2026 **SHOPPERAI, AGENTIC AI SHOPPING ASSISTANT** **Google ADK, Gemini, GCP**

- **Architecture:** Shipped a production agentic shopping assistant with deterministic, stage-based workflows for product discovery and virtual try-on, deployed on GCP Cloud Run.
- **Reliability:** Reduced hallucinated outputs and improved response relevance through structured prompt engineering, tool-use constraints, and grounded retrieval inside the Google ADK framework.

2024 **RAG DOCUMENT INTELLIGENCE SYSTEM** **LangChain, ChromaDB, SBERT**

- **RAG Pipeline:** Designed a RAG pipeline over 1,000+ documents using SBERT and ChromaDB (92% retrieval accuracy, 18% output-quality gain), deployed on GCP with async FastAPI endpoints sustaining sub-500ms latency and 99.5% uptime under load testing.

EDUCATION

Aug 2024 - May 2026 **UNIVERSITY OF HOUSTON** **HOUSTON, TX**
M.S. Data Science, GPA 3.9/4.00.

Aug 2020 - May 2024 **AMRITA VISHWA VIDYAPEETHAM** **BENGALURU, INDIA**
B.Tech, Electronics and Communication Engineering, GPA 3.50/4.00.

LEADERSHIP

Oct 2020 - Jun 2024 **CLASS REPRESENTATIVE, AMRITA VISHWA VIDYAPEETHAM** **BENGALURU, INDIA**
Served all four years; liaised between faculty and 60+ students, organized events, guided peers.

HONORS

4th place / 200 teams, ODSC AI x Google Cloud Agentic AI Hackathon (Dec 2025).

1st place / 720 students, IIT Tirupati senior-level Talent Search Examination (2019).